

Data Flow Diagram

Date	15 July 2026
Team ID	XXXXXX
Project Name	FloodGuard - AI/ML Flood Prediction System
Maximum Marks	2 Marks

DFD Symbol Legend

Symbol	Name	Description
Oval	External Entity	User / Browser submitting weather data
Rectangle (numbered)	Process	Validate Input, Scale Features, Run Model, Log Result
Rectangle (solid)	Data Store	flood_dataset.xlsx, flood_model.joblib, SQLite DB
Labeled Arrow	Data Flow	Movement of data between the above (e.g., "weather values")

Level-1 Data Flow (FloodGuard)

- User (External Entity) → submits weather form values → [1.0 Validate Input]
- [1.0 Validate Input] → clean numeric values → [2.0 Scale Features] (using scaler.joblib)
- [2.0 Scale Features] → scaled feature vector → [3.0 Run Prediction Model] (flood_model.joblib)
- [3.0 Run Prediction Model] → prediction + probability → [4.0 Log Result to Database]
- [4.0 Log Result to Database] → stores record → SQLite Data Store (users, weather_data, ml_models, prediction_results)
- [4.0 Log Result to Database] → JSON response → User (rendered as risk badge + probability bars)

FLOOD PREDICTION SYSTEM – DATA FLOW DIAGRAM (LEVEL 1)

